Terraform Tasks - 01

- Install Terraform on your PC.

```
PS C:\Users\Viqaas\Desktop\Testing\Terraform> terraform -v
Terraform v1.14.6
on windows_amd64
PS C:\Users\Viqaas\Desktop\Testing\Terraform>
```

- Execute all the templates shown in the video.

```
PS C:\Users\Viqaas\Desktop\Testing\Terraform> terraform init
Initializing the backend...
Initializing provider plugins...
- Finding latest version of hashicorp/local...
- Installing hashicorp/local v2.7.0...
- Installed hashicorp/local v2.7.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
```

```
Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

local_file.my_pet: Creating...
local_file.my_pet: Creation complete after 0s [id=7e4db4fbfdbb108bdd04692602bae3e9bd1e1b68]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
```

main.tf | pets.txt

pets.txt

```
1   I love pets!
```

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```
Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
PS C:\Users\Viqaas\Desktop\Testing\Terraform> terraform plan
local_file.my_pet: Refreshing state... [id=7e4db4fbfdbb108bdd04692602bae3e9bd1e1b68]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated wi
-/+ destroy and then create replacement

Terraform will perform the following actions:

  # local_file.my_pet must be replaced
-/+ resource "local_file" "my_pet" {
      ~ content              = "I love pets!" -> "I adore pets!" # forces replacement
      ~ content_base64sha256 = "4LBQ/UV/Gxp6DWNEp+upXEfPrTTpUCpfiAkOpJixJEc=" -> (known after apply)
      ~ content_base64sha512 = "wKb93aeKXXwWVvqgU6yfOuiKC6dLDLLWp7dclsVtlgenVx4/BUpKiEnsdA6EbHgUmSDTp/+Si9wFeXH8/
      ~ content_md5          = "3dd08189b6cbd661de6b25ed369f5746" -> (known after apply)
      ~ content_sha1         = "7e4db4fbfdbb108bdd04692602bae3e9bd1e1b68" -> (known after apply)
      ~ content_sha256       = "e0b050fd457f1b1a7a0d6344a7eba95c47cfad34e9502a5f88090ea498b12447" -> (known after
      ~ content_sha512       = "c0a6fddda78a5d7c1656faa053ac9f3ae88a0ba74b0cb2d6a7b75c96c56d9607a7571e3f054a4a884
r apply)
```

-

```
Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

local_file.my_pet: Destroying... [id=7e4db4fbfdbb108bdd04692602bae3e9bd1e1b68]
local_file.my_pet: Destruction complete after 0s
local_file.my_pet: Creating...
local_file.my_pet: Creation complete after 0s [id=51748dfda322b061423085a5abe69db47dbd69a7]

Apply complete! Resources: 1 added, 0 changed, 1 destroyed.
```

main.tf pets.txt

pets.txt

```
1   I adore pets!
```

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main.tf > resource "random_pet" "my-pet"

```
resource "local_file" "my_pet" {
  filename = "pets.txt"
  content  = "I love pets!"
}

resource "random_pet" "my-pet" {
  prefix    = "MR"
  separator = "."
  length    = 1
}
```

-

```
PS C:\Users\Viqaas\Desktop\Testing\Terraform> terraform fmt
main.tf
PS C:\Users\Viqaas\Desktop\Testing\Terraform> terraform init
Initializing the backend...
Initializing provider plugins...
- Reusing previous version of hashicorp/local from the dependency lock file
- Reusing previous version of hashicorp/random from the dependency lock file
- Using previously-installed hashicorp/local v2.7.0
- Using previously-installed hashicorp/random v3.8.1

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
```

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```
local_file.my_pet: Destroying... [id=51748dfda322b061423085a5abe69db47dbd69a7]
random_pet.my-pet: Creating...
local_file.my_pet: Destruction complete after 0s
random_pet.my-pet: Creation complete after 0s [id=MR.giraffe]
local_file.my_pet: Creating...
local_file.my_pet: Creation complete after 0s [id=7e4db4fbfdbb108bdd04692602bae3e9bd1e1b68]

Apply complete! Resources: 2 added, 0 changed, 1 destroyed.
```


```
{
  "schema_version": 0,
  "attributes": {
    "id": "MR.giraffe",
    "keepers": null,
    "length": 1,
    "prefix": "MR",
    "separator": "."
  },
  "sensitive_attributes": [],
```

```
"instances": [
  {
    "schema_version": 0,
    "attributes": {
      "id": "Miss.maggot",
      "keepers": null,
      "length": 1,
      "prefix": "Miss",
      "separator": "."
    },
    "sensitive_attributes": [],
```


- Note below points,
- Terraform Init: Prepares folder and downloads plugins. Creates .terraform folder.
  - terraform init -upgrade: Updates providers/modules.
  - terraform init -reconfigure: Re-initializes setup.
- Terraform Plan: Shows what Terraform will create/change (preview).
  - terraform plan -out=tfplan: Saves plan to a file.
  - terraform plan -var="name=value": Pass variable value.
- Terraform Apply: Actually creates the resources.
  - terraform apply -auto-approve: Skip confirmation.
  - terraform apply tfplan: Apply saved plan file.
- Terraform Provider: It is the plugin that connects Terraform to a cloud or service.

Terraform Tasks - 01

- Integrate a sample Terraform template in Jenkins.

```
[1mPlan:[0m 1 to add, 0 to change, 0 to destroy.
[0m[1mlocal_file.demo: Creating...[0m[0m
[0m[1mlocal_file.demo: Creation complete after 0s [id=7f9f4ed5f0ea724ebdb79204e5e07f70b034b88b][0m
[0m[1m[32m
Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
[0m
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] slackSend
Slack Send Pipeline step running, values are - baseUrl: <empty>, teamDomain: techiehorizon, channel: #jenkins-integration, color: <empty>,
notifyCommitters: false, iconEmoji: <empty>, username: <empty>, timestamp: <empty>
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

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```
jenkins@634d24255021:~/workspace/Terraform$ ls
DevOps  Kill  Lie  Line  README.md  Sample  Testing.tf  Why  fighter  file.txt  hello.txt  killer  password  terraform.tfstate  test.txt
jenkins@634d24255021:~/workspace/Terraform$
```

-


```
  "resources": [
    {
      "mode": "managed",
      "type": "local_file",
      "name": "demo",
      "provider": "provider[\"registry.terraform.io/hashicorp/local\"]",
      "instances": [
        {
          "schema_version": 0,
          "attributes": {
            "content": "Hello from Terraform \u0026 Jenkins",
            "content_base64": null,
            "content_base64sha256": "aVDv+4YrOi8epTGRLT5hS8UPVUa141wcwSVOGHuhmtI="
            "content_base64sha512": "o7NF++q5vAoMoLhGkZSsNJJ6cPhThqgF+KZ7XxqGBwaAS
            "content_md5": "225b3d5b0b34f671869f77eec4f27701",
            "content_sha1": "7f9f4ed5f0ea724ebdb79204e5e07f70b034b88b",
            "content_sha256": "6950effb862b3a2f1ea531912d3e614bc50f5546b5e35c1cc12
            "content_sha512": "a3b345fbeab9bc0a0ca0b8469194ac34927a70f85386a805f8a
            "directory_permission": "0777",
            "file_permission": "0777",
            "filename": "hello.txt",
            "id": "7f9f4ed5f0ea724ebdb79204e5e07f70b034b88b",
            "sensitive_content": null,
            "source": null
          },
          "sensitive_attributes": []
        }
      ]
    }
  ],
  "check_results": null
}
jenkins@634d24255021:~/workspace/Terraform$
```